

Phase 4: Assembly and Prototype Report

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NAPKN: Natalie Huang, Kaylyn Nguyen, Pranaav Dev Muneeswaran, Ashley Roselynn

Vincent, Noah Chie

University of California, Irvine

Existing research and objective

Rheumatoid arthritis (RA) is a chronic autoimmune disease that causes chronic joint inflammation. As of 2019, over 18 million people worldwide have been diagnosed with RA which is just a subset of the 52 million total patients in the United States. This displays a huge need that needs to be met. These patients often self-administer medication weekly or even daily through subcutaneous (into fatty tissue) injections, which prior research has shown to require hand movements and positioning that cause pain [2] and treatment non-compliance [3]. Patients in advanced stages of RA struggle with firmly gripping a syringe, pressing buttons or plungers, and stabilizing a syringe for ten seconds, the time required to inject the medication. Since these medications need to be taken at regular intervals, these issues ultimately lead to further pain, improper dosing, and, in many cases, patients that skip treatment completely (nonadherence).

RA has many treatments, the majority of which involve patients regularly injecting biological medications subcutaneously. Today's auto-injectors are designed for people with strong, steady hands, so there is much room for improvement in accommodating the dexterity and weak joints of RA patients when it comes to creating auto-injector devices.

Humira Pen (adalimumab)

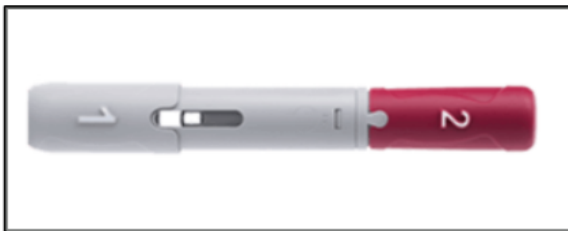


Figure 1. Humira Pen by *Abbvie*

Abbvie designed a similar autoinjector for patients with RA, the Humira Pen (adalimumab), which is only administered for one use, and utilizes a push-button [6].

However, this device requires a strong pull due to the small barrel size while also requiring a

button press that can be difficult for RA patients with tremors, considering the need to hold for

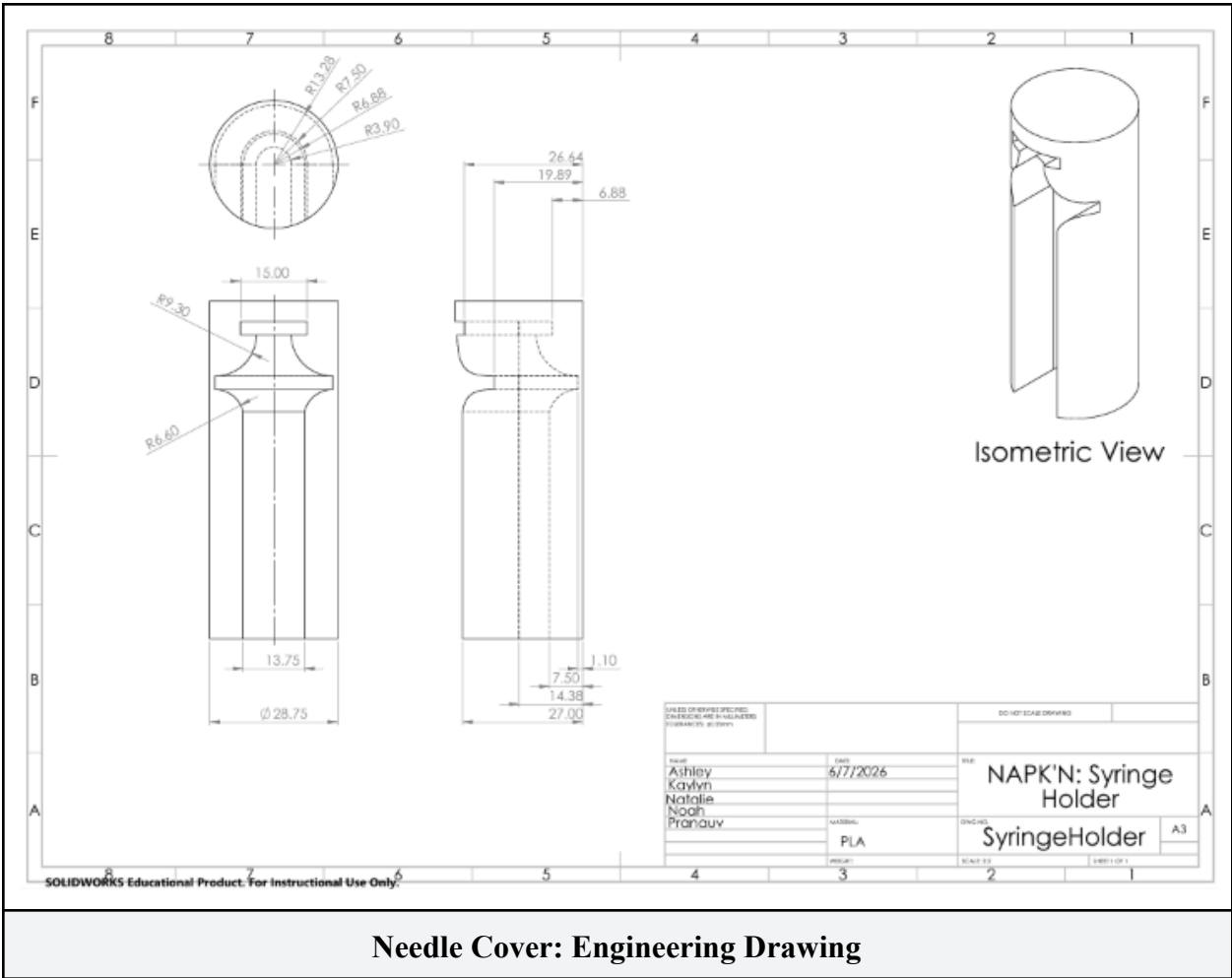
10 seconds (Figure 1). Another similar device, *Enbrel's* SureClick autoinjector, is a single button, subcutaneous pen requiring the removal of a cap through a twisting force that can cause shaking of the hands [7]. Thus, we sought to create an ergonomic autoinjector device that improves upon the current issues that RA patients face with autoinjector treatments to allow better grip and remove any buttons or twisting mechanisms that may be difficult to navigate for those with dexterity issues. With this project, we hoped to improve the comfort, control, and adherence of medication administration for rheumatoid arthritis patients. Our product aims to address these drawbacks present in current autoinjectors to accommodate the discussed current market.

This product also aims to address this issue from a financial standpoint. The variety pack of springs costs around \$15.99, with the main large spring costing \$13.98. The 5 mL syringe used to carry the medication costs around \$11.98 for a pack of 10 with 2 total units purchased. Though costs for additional medication will add to this value, this would be less than the overall costs of purchasing current injectors, which traditionally cost from around \$33 up to \$133, every time a treatment is needed.

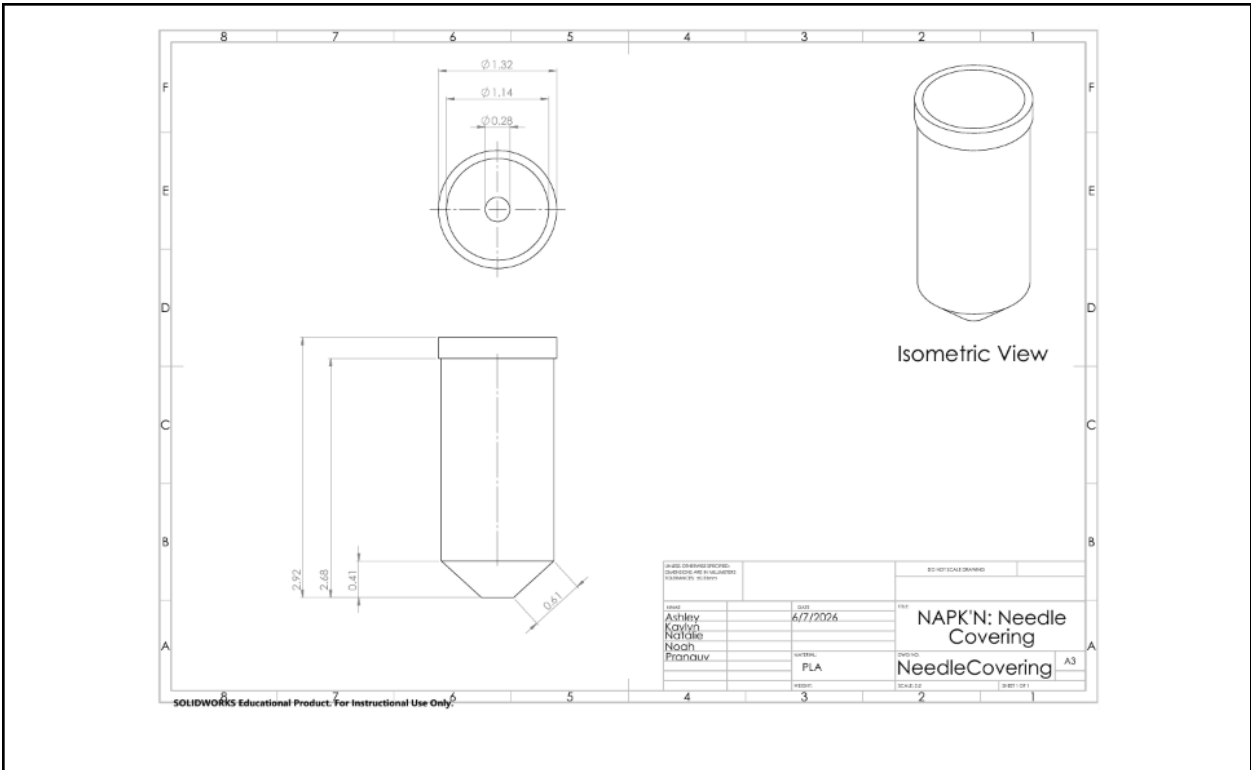
Our device utilizes three moving mechanisms, all of which share the common goal of allowing the user to keep their hand steady and minimize the usage of the prominent joints of interest. In addition, our devices emphasize reusability in that the medication capsule itself is removable, in which the appropriate medicine that needs to be injected will be stored. Additionally, the next steps in this project involve improving the ergonomics of the grip itself, as well as expanding the target audience to patients with similar conditions to rheumatoid arthritis.

Part Design

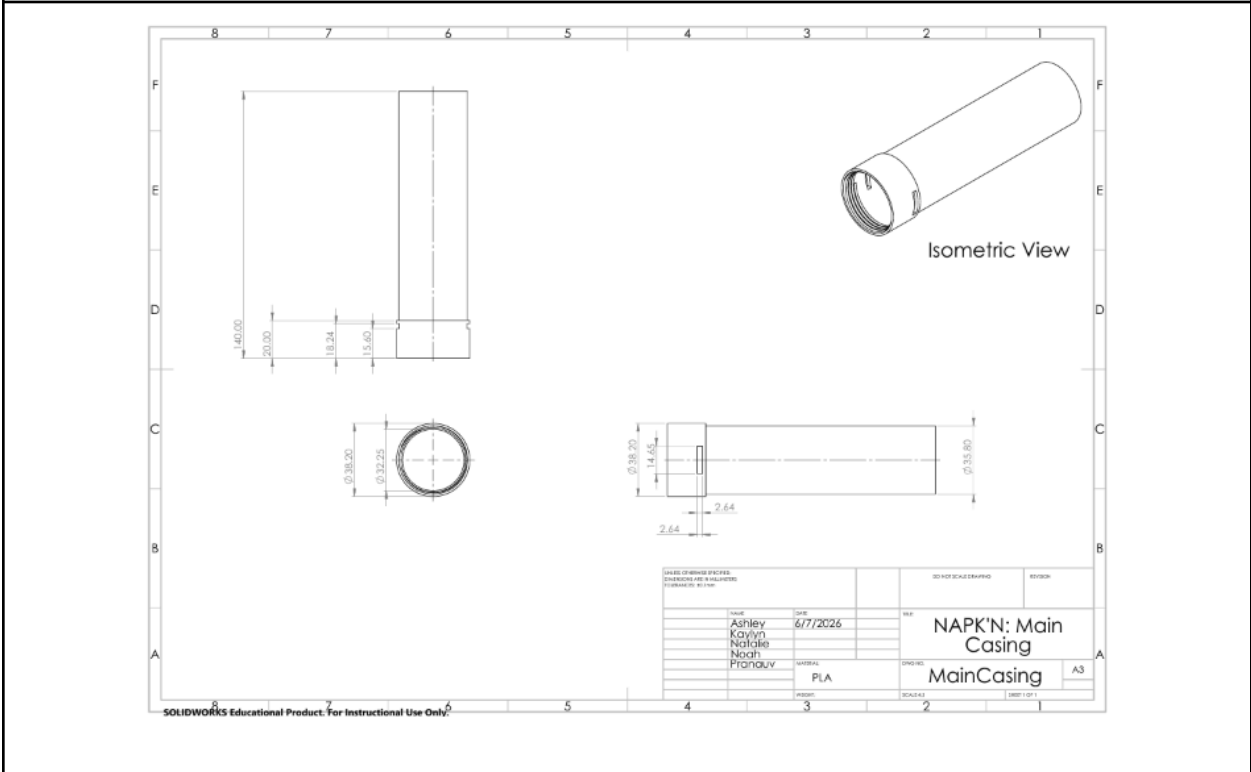
Part Design: Design Videos
• Syringe Holder • Needle Cover • Main Casing • Grip • Cap • Shim
Syringe Holder: Engineering Drawing



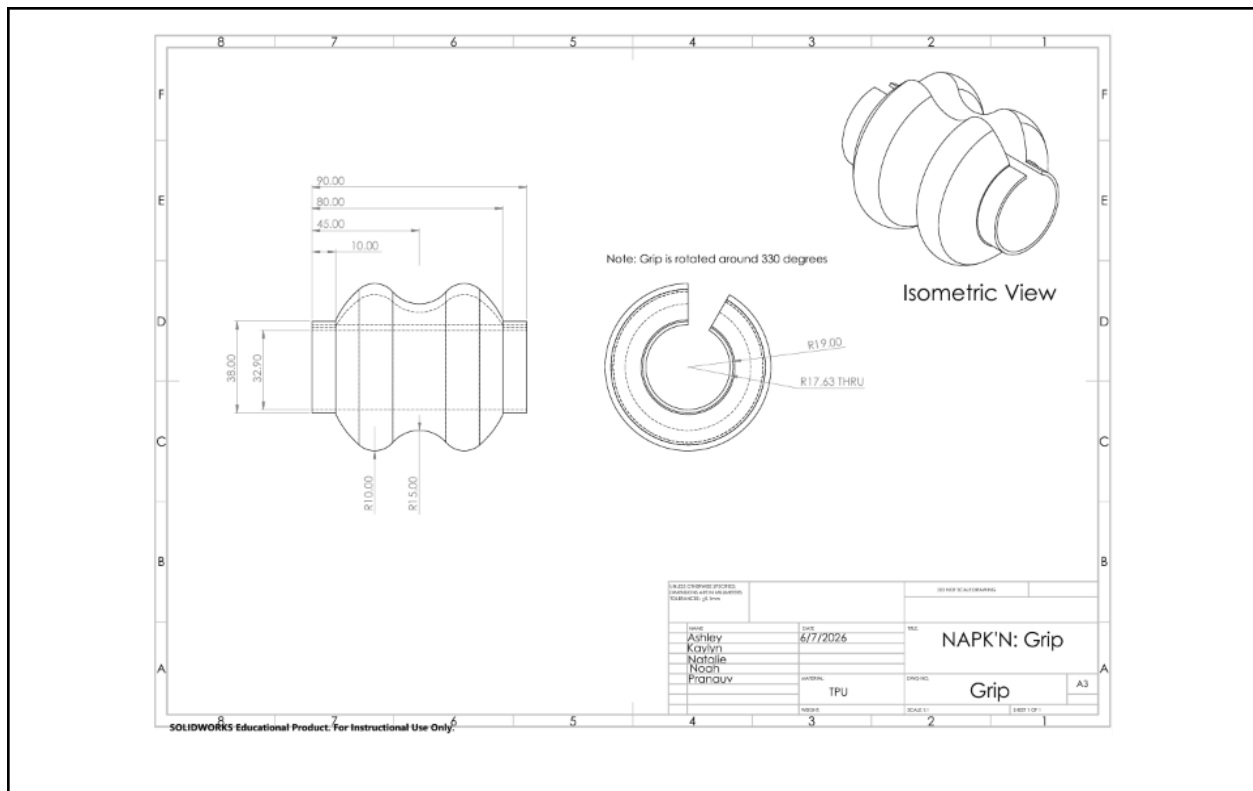
Needle Cover: Engineering Drawing



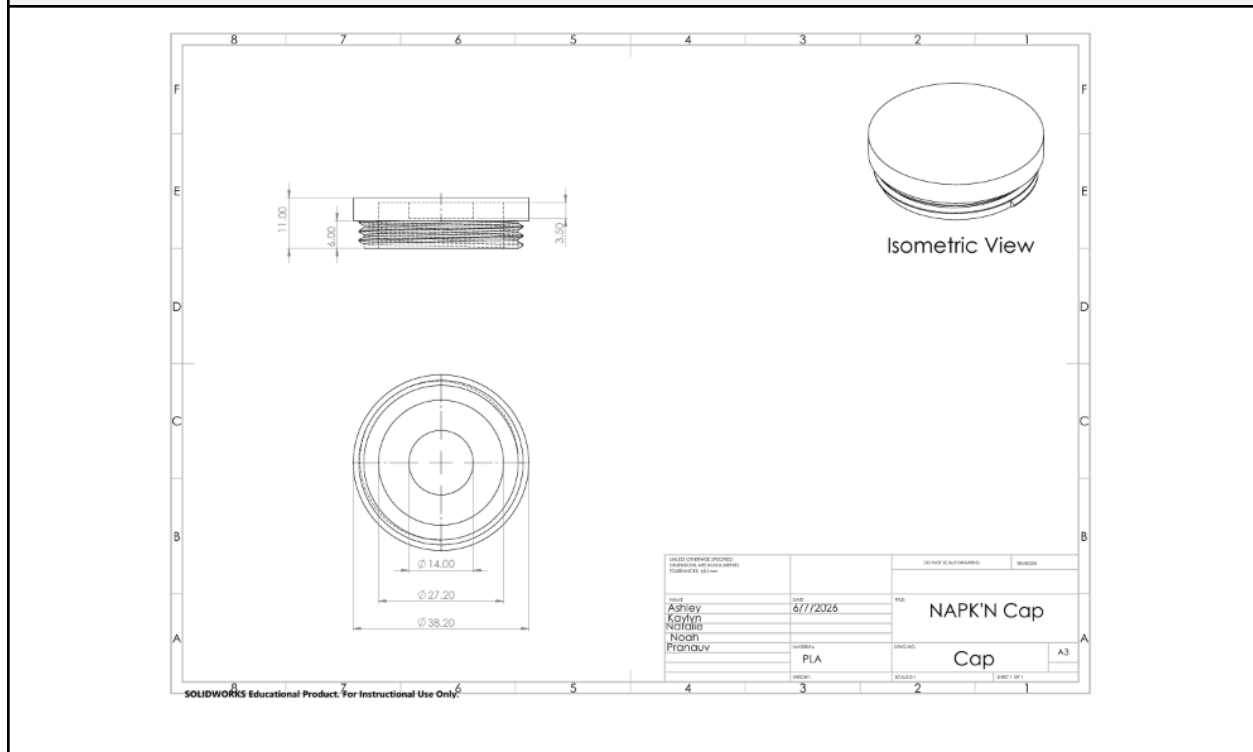
Main Casing: Engineering Drawing



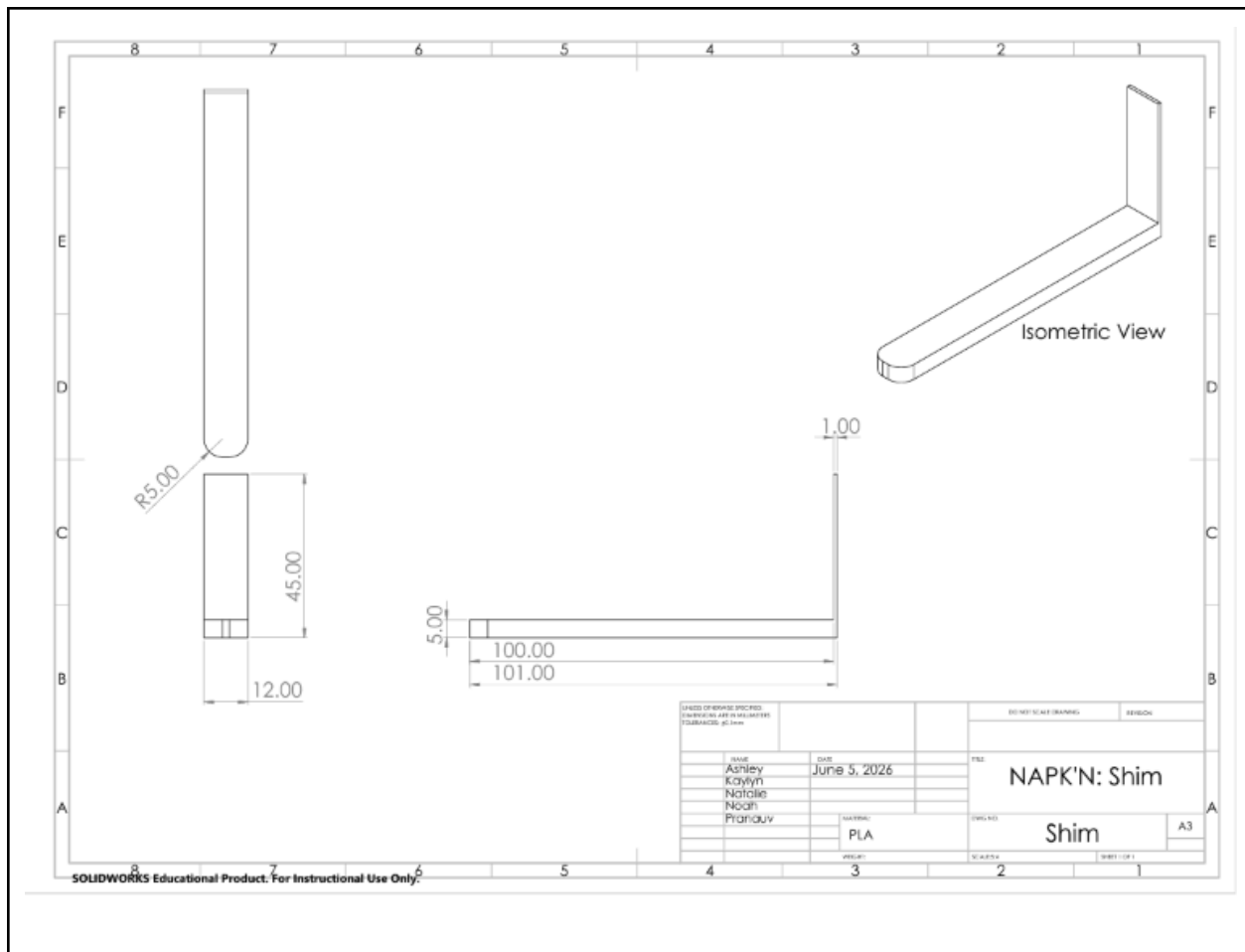
Grip: Engineering Drawing



Housing Cap: Engineering Drawing



Shim: Engineering Drawing



Assembly Design

Assembly Description YouTube Link: [Click Here](#)

Assembly Engineering Drawing with Exploded View:

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1

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ITEM NO.	PART NUMBER	DESCRIPTION	MATERIAL	QTY.
1	PartDvFINAL	Main Casing Cap	PLA Plastic	1
2	Shim	Spring Lock	TPU Plastic	1
3	Copy of CompletedGrip_NAPKN (5) (5) (1) (1) (2) (1)	Ambidextrous Grip	PLA Plastic	1
4	Copy of SyringeHolder	Syringe Casing	PLA Plastic	1
5	PartMainCasingvFINAL	Main Casing	PLA Plastic	1
6	NeedleCoveringvFINAL	Needle Cover	PLA Plastic	1

PROPRIETARY AND CONFIDENTIAL

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UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN MM/MS

TOLERANCES:

FRICTION ± 0.1 mm

SLIDING ± 0.25 mm

INTERPRET GEOMETRIC TOLERANCING PER:

MATERIAL: View BOM

FINISH:

DO NOT SCALE DRAWING

NAME	DATE
Contributors	Noah 6/7/2026
	Natalie
	Kaylyn
	Pranav
	Ashley

COMMENTS:

SolidWorks does not have PLA plastic installed.

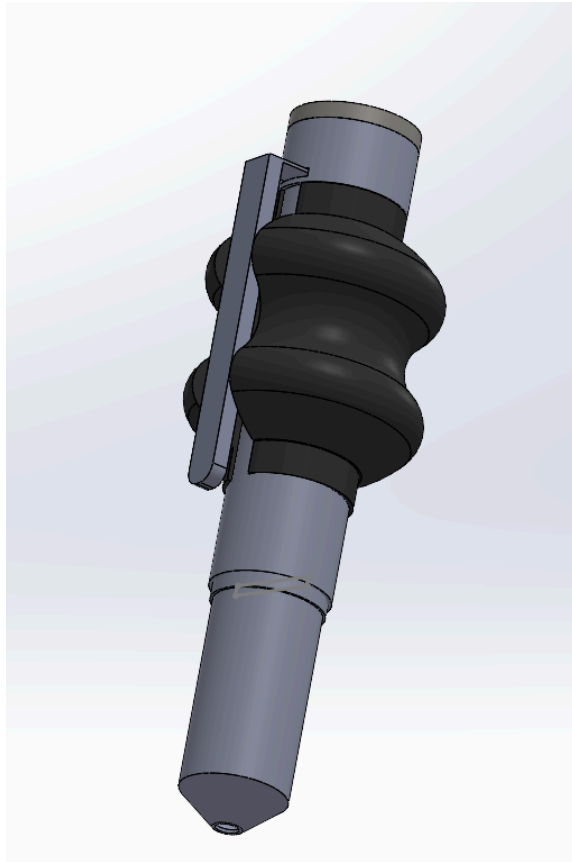
Springs are not part of our SolidWorks Assembly. Unable to calculate the total weight.

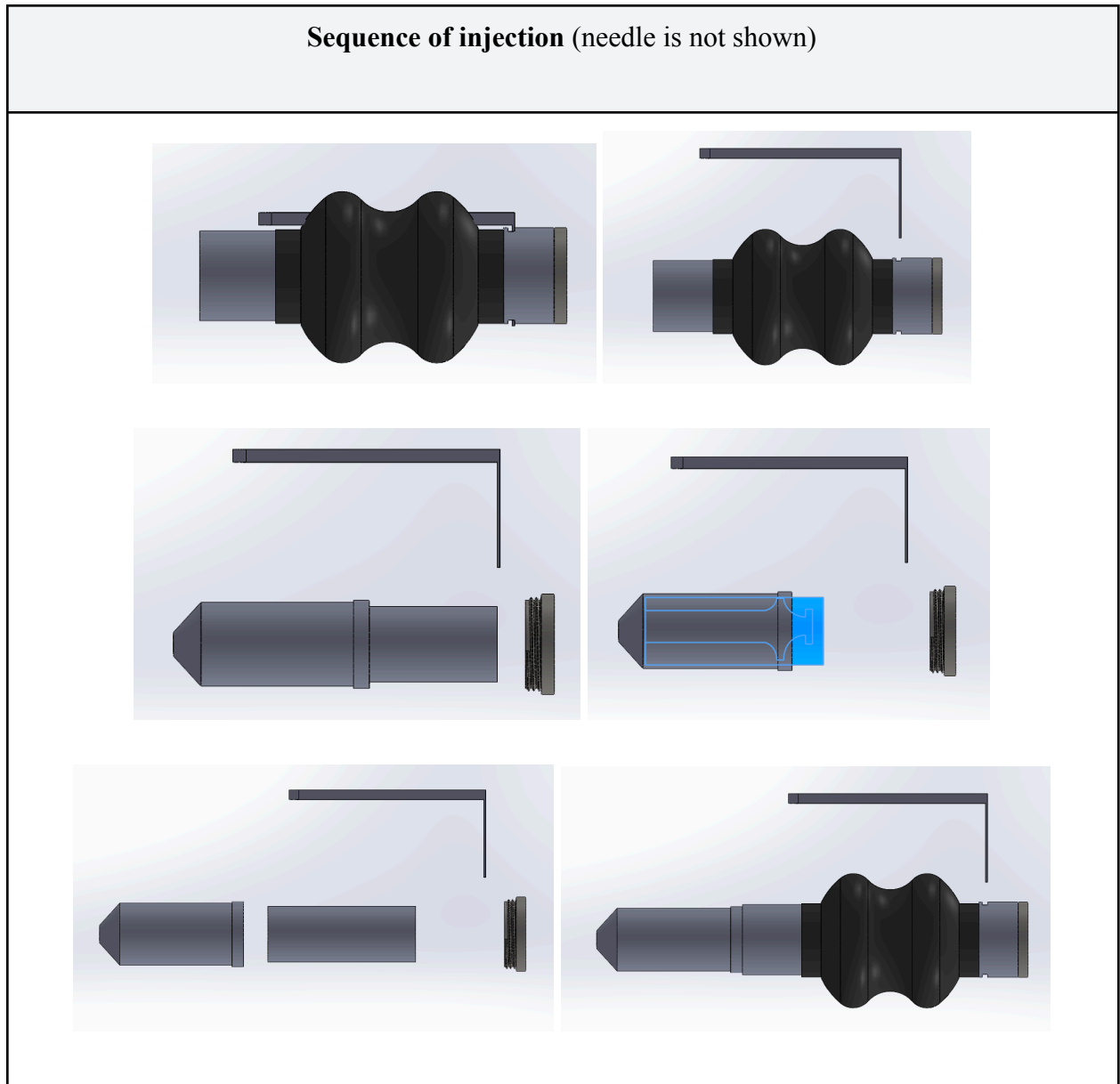
TITLE: **NAPK'N Full Assembly**

SIZE	DWG. NO.	REV
A	Assem1	

SCALE: 1:2 WEIGHT: N/A SHEET 1 OF 1

Our Device



Sequence of injection (needle is not shown)

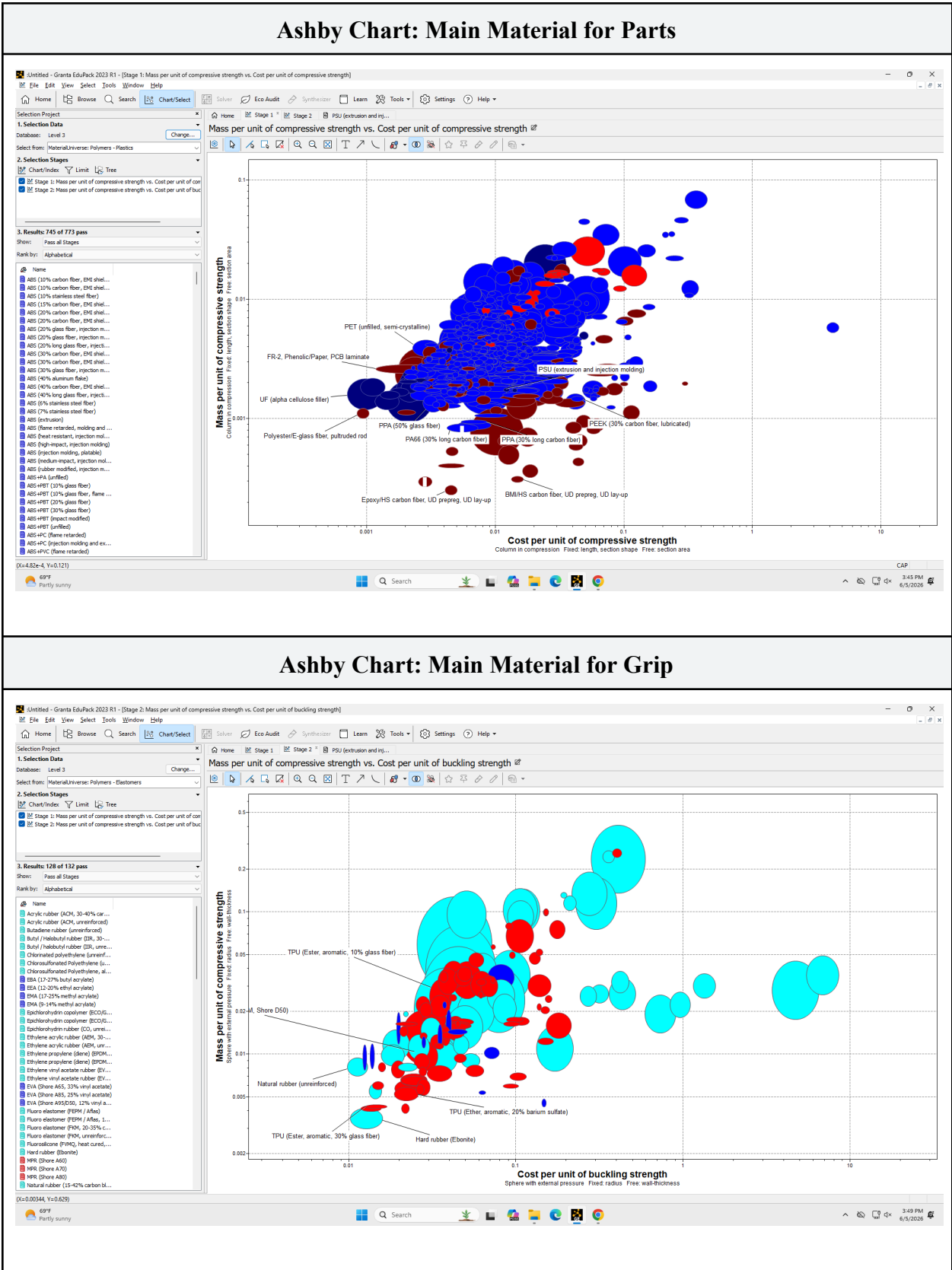
- **The design incorporates three moving parts:**
 - Needle locking mechanism with spring and rod
 - The needle is injected
 - Once released, the needle cover is deployed

Materials and prototyping design

If given the opportunity to use any given material for our prototype, we would utilize the Ashby chart to determine which materials we would select. Depending on which aspects are selected to optimize and prioritize for the success of the project, the parameters will produce various material options on a single graph to choose from.

Regarding every part of the prototype except the grip, we wanted to prioritize the mass of the material and the cost of manufacturing. This was done in order to optimize for a material that is strong enough for all the parts that will encase the syringe and be utilized for the working mechanism, while being lightweight for patient usage. It needs to be sturdy enough to support the strength of the springs and the force of the needle injection, while being light enough to allow for patient comfort during usage. We also want a low-cost material to help with mass production. From the generated Ashby chart, we determine that polysulfone (PSU) is an ideal material for the prototype since it meets the expectations listed earlier.

In comparison, for the grip component, we decided to use a different material in comparison to the rest of the project since it serves a different purpose; it is meant to add additional comfort for the patient's usage. We want a material that is on the cushiony side to allow the patient to fully grasp & mold their fingers into the grip to fully maximize comfort. When generating the Ashby chart, we prioritized the mass of the product and its performance under buckling strength since the patient will be grasping it. We wanted the material to be lightweight, while being able to undergo the force from the patient's grasp. From the generated Ashby chart, we determine that TPU 30% is an ideal material to use for production of the grip.



Instructions on Building Prototype		
1	Insert the syringe with needle into the syringe holder	
2	Place three small diameter “weak” springs on the needle, ensuring the entire needle is covered through the springs and use super glue to ensure they stay stable	

3	Superglue two large diameter “strong” springs using a washer to have the spring be an optimal length and superglue the spring onto the housing cap (the cap is designed to have a cylinder that the spring sits flesh on, the superglue is to ensure stability)	
4	Insert the syringe holder into the needle cover	
5	Insert the needle cover along with the syringe holder into the main body of the housing	

6	Screw the housing cap that now has the spring attached to it onto the main housing	
7	Compress the large spring and lock the mechanism in the loaded position using a shim (we had to use a metal file as a shim as the shim we printed from PLA was not strong enough)	
8	Slide the grip onto the outside of the main housing so it sits flush with the upper groove	

Cost analysis and results

Vendor 1: Amazon					
Quantity	Vendor	Description	Item No.	Link	Total
1	Amazon	Spring <i>Variety pack with multiple lengths</i>	B0BBQL2SN 3	https://a.co/d/0ht7xBk1	\$15.99
2	Amazon	Large Spring	B008RG48JI	https://a.co/d/0fGTO1Ly	\$13.98
1	Amazon	Syringe - 2mL	B0B95ZVVG	https://a.co/d/0iFEBPFI	\$8.99
2	Amazon	Syringe -	B0DSBLCSD	https://a.co/d/	\$11.98

		5mL	Q	02DmOqCD	
1	Amazon	Syringe - 10mL	B0CKYHNGH3	https://a.co/d/06Xrto3Y	\$11.99
Total without tax:					\$62.93

➤ Green = Parts that were used for the final prototype.

Total 3-D printing cost over 5 iterations: \$121.52 (20.253 hours)

Cost of manufacturing 1 unit (without 3D printing) = \$14.67

Cost of manufacturing 1 unit (with 3D printing [approx 4 hours for 1 iteration]) = \$38.67

The parts that we purchased were necessary to match the conditions of the intended user. The syringes were purchased to imitate the current self-injection devices that are currently on the market. We also purchased springs, which were necessary to design and test the moving parts of our product.

Ensuring the fitting of the spring to the syringe was difficult just based on the images and dimensions found online. Therefore, we purchased different sizes of springs in bulk and syringes with the goal of iterating multiple times to find what sizes suit the product the best. As shown above, only three of the five components that were initially purchased were used in the final design, which reduced our costs significantly.

Total Money Spent: \$184.45

Overall, we stayed below the \$215 budget and were able to reduce the final costs of our device since the beginning of our prototyping.

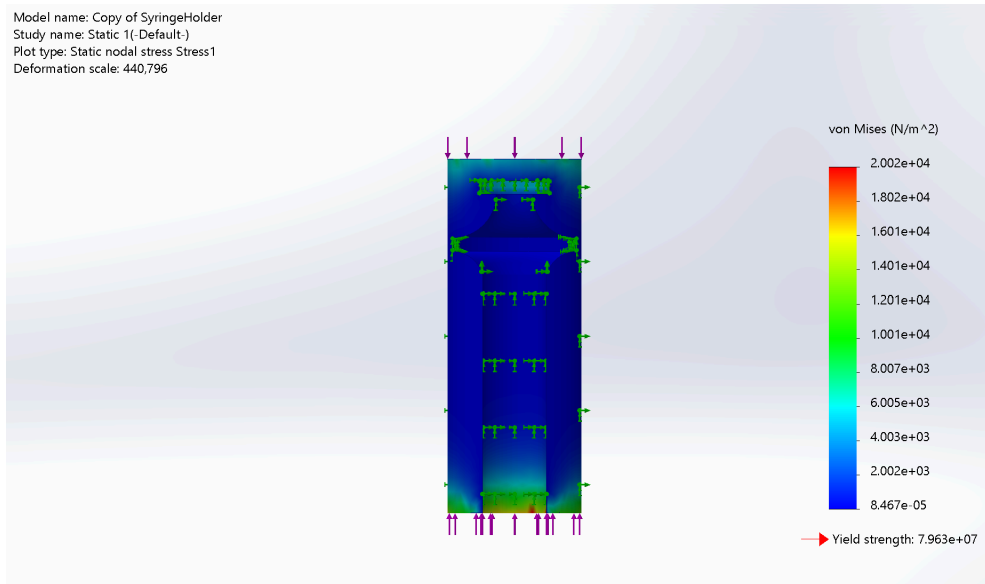
Iterative testing method and results

We decided to test the parts individually by simulating the forces that would be applied to them in order to view the structural integrity of the parts more clearly. All of the parts are meant to be PSU, as determined by the Ashby Charts, except for the grip, which is TPU. We manually added these materials as well as their material properties as they do not have templates in SolidWorks.

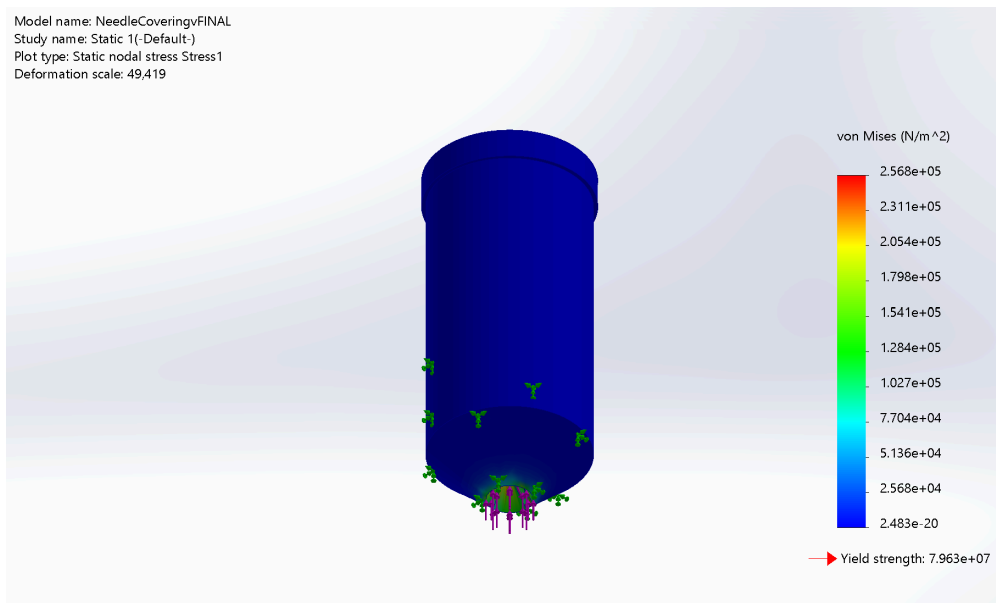
We compared the maximum stress to the yield strength, and considered the part within tolerable limits if the maximum stress was less than the yield strength. These tests showed that the stresses that would be applied to the parts when assembled and used are all within tolerable limits. This allows us to confidently state that the parts will be able to withstand the forces that would be applied to them during usage. These tests showed that the stresses that would be applied to the parts when assembled and used are all within tolerable limits.

Part	Syringe Holder	Needle Cover	Main Casing	Cap	Shim	Grip
Force Applied (N)	5	5	1	5	1	10
Yield Strength (N/m ²)	7.963×10^7	7.963×10^7	7.963×10^7	7.963×10^7	7.963×10^7	1.169×10^8
Max Stress (N/m ²)	2.002×10^4	2.568×10^5	2.138×10^4	4.811×10^5	8.092×10^6	2.139×10^3
Within Limits?	Yes	Yes	Yes	Yes	Yes	Yes

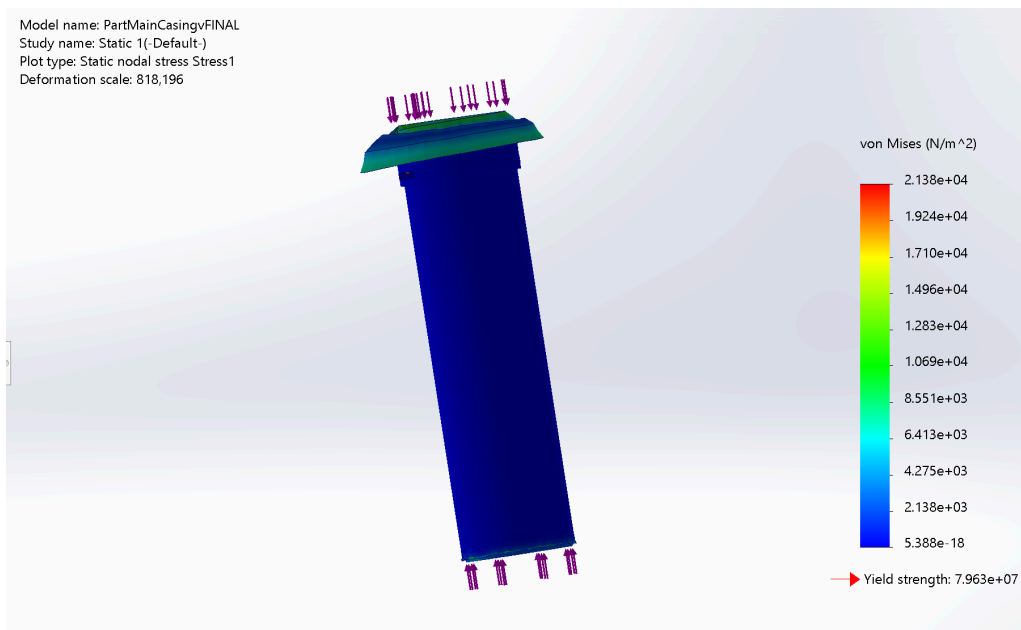
For the syringe holder, we tested stress with deformation when a force of 5N was applied from the top and the bottom to simulate the compression from the springs. We applied a fixture to the outer surface, which would be held up by the outer casing, and the surfaces in contact with the syringe.



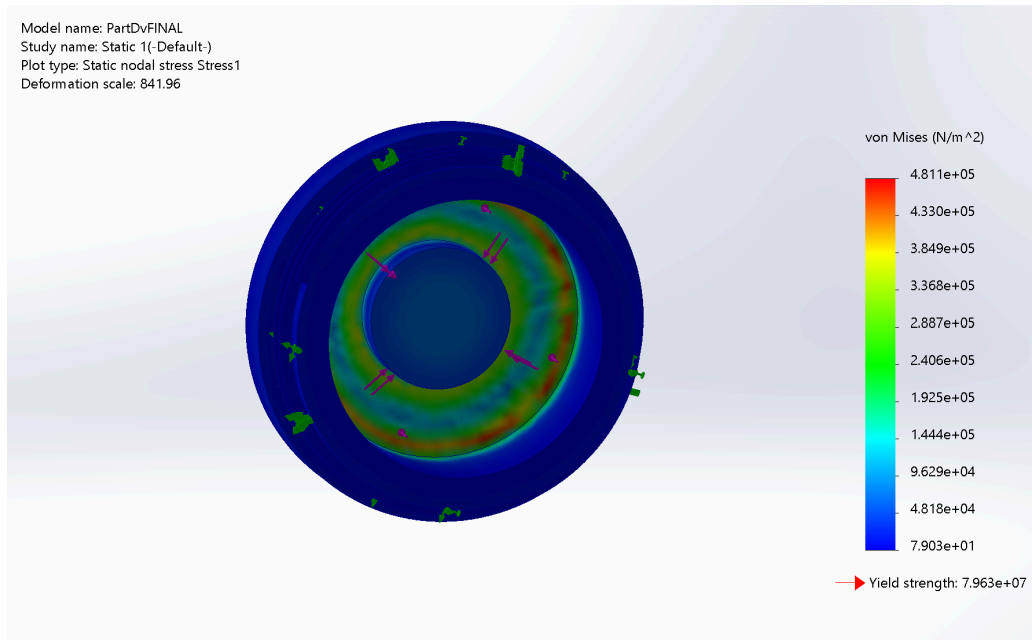
For the needle covering, we simulated a force pressing up as though the device were pressing against skin while a second force pushed down from the inside of the needle covering to simulate the force from the springs. Both forces were assumed to be 5N with a fixture applied to the cylindrical surface that would be pressed against the outer casing and the syringe holder.



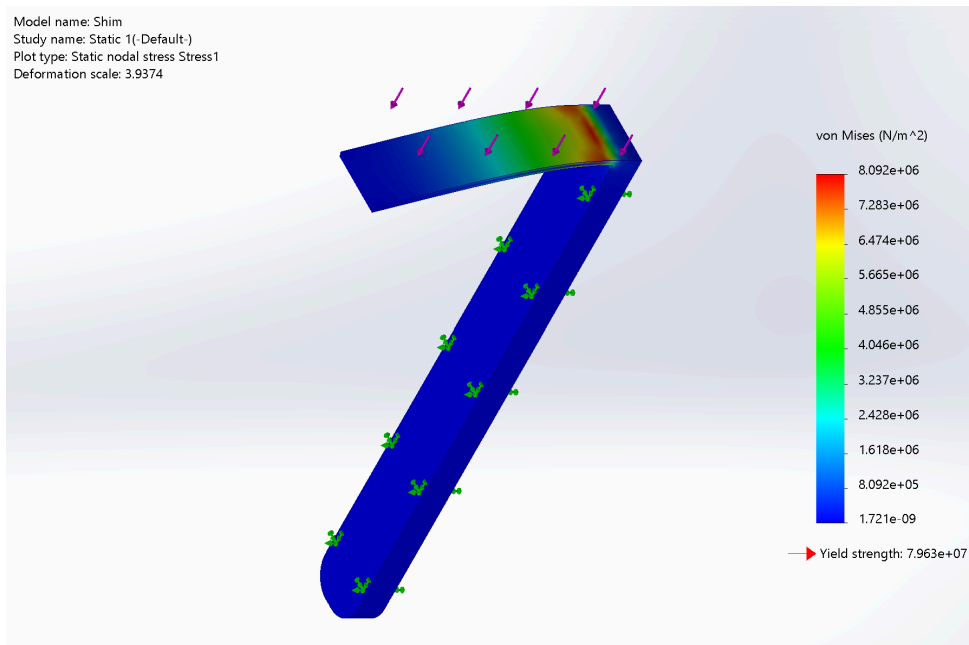
For the main casing, we tested the vertical compressive forces that would occur during the injection such as the pressure from the cap and from the skin. This was assumed to be 1N, since there are no spring interactions. We placed fixtures on the interior surface.



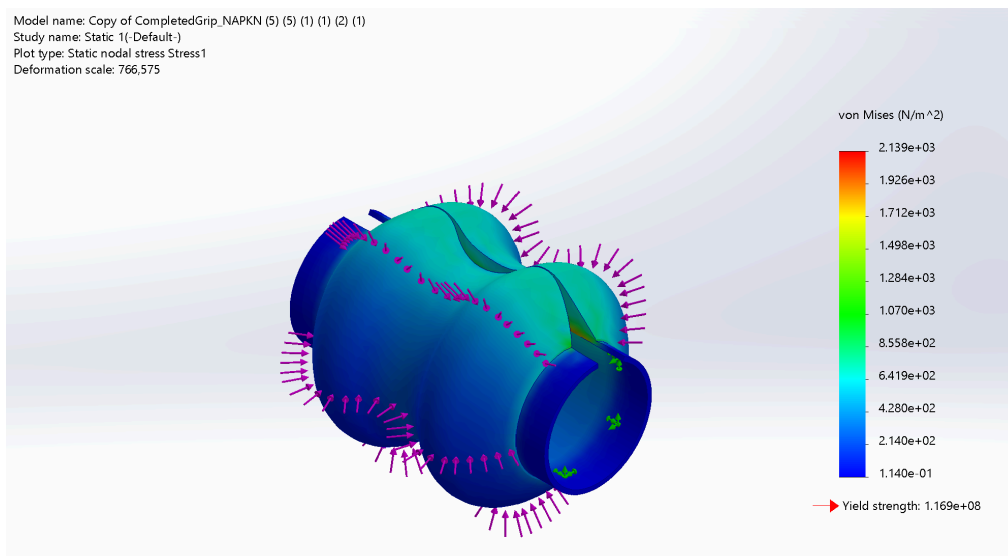
For the cap, we tested the stress from the forces of the spring at 5N with a fixture on the threaded surface. The spring is supposed to be attached to the interior of the cap, around the extruded circle.



For the shim, we tested a force of 1N pressed down onto the top, with a fixture applied to the sides of the handle. While this is still within acceptable parameters, the maximum stress is a lot closer to the yield strength than for our other parts, so a thicker and more durable part may prove to be useful for future iterations.



For the grip, we tested the stress with deformation when compressed from all directions at 100N, to simulate someone squeezing on it with a lot of force. We applied a fixture to the interior cylindrical surface of the grip.



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